ECDLP research program: state, leads and next decisions

30 October 2026, 15:21 UTC snapshot. Evidence assessment and source-level investigation; no new algorithm, theorem, experiment, benchmark or challenge solve. The renewed request is to assess the program and keep pursuing leads.

Assessment: code availability is no longer the central hashed-ECDL gap. The priorities are independent validation of the general contraction route and complete-cost evaluation of the finite algorithms already delivered. Generic engineering, algebraic solving and special-family transfers remain separate tracks. Progress in one does not establish the others.

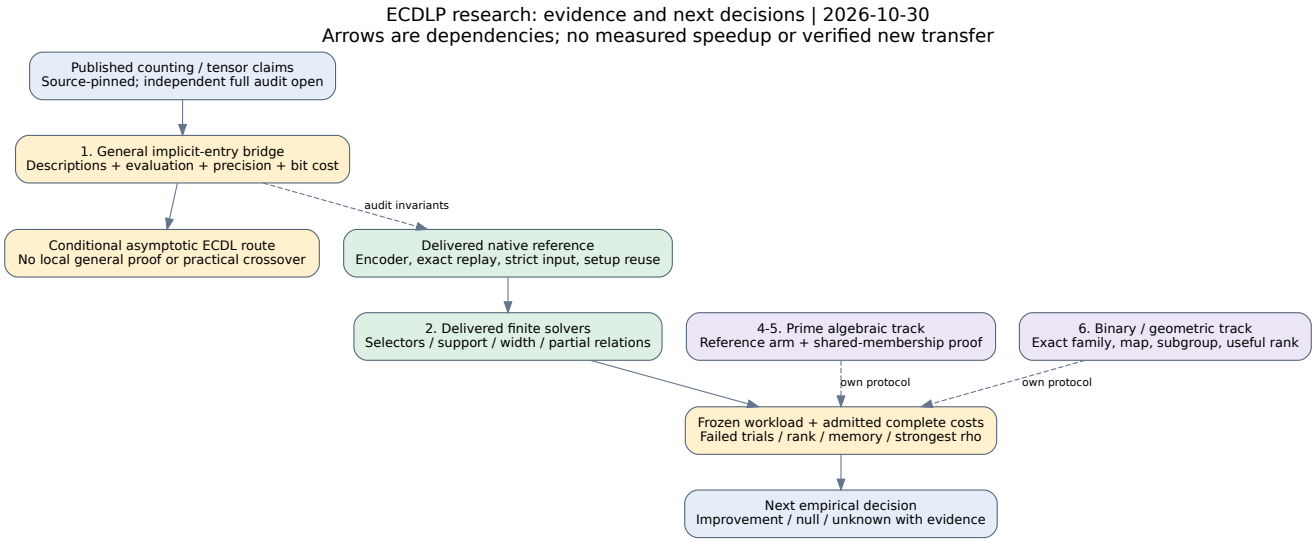


Figure 1: Evidence dependencies; no timing data.

Snapshot and method

Reviewed crypto main fc9af16fa1ce0654ee9df3ce63101c487ed215fb, cryptanalysis main 5939def096fd181973e1abb7279644a8aa531557, and atlas 58f41b1625c66c637b0bf2b373b5cf65c17cf85c. This is a targeted assessment, not an exhaustive audit of all repository experiments. Merged PRs were not reopened or put back into tracking.

Read canonical problems, implementation receipts, leaderboard qualifications and atlas matches before searching current primary sources; then inspect native admission/search code and actual formalization entry points. Pins and hashes are in sources.json (sources.json). The bridge map (BRIDGE.md) records the first pursued lead and remaining obligations. The PDF (report.pdf), editable diagram (dependency.dot) and renderer (render_report.py) accompany it.

The old local atlas brief reported 150 leads. Reading current pinned records gives 152: 52 open with a next step, 17 blocked, 9 unexamined, 74 in closed or exhausted categories. These are metadata counts, not probabilities or a complete literature census. Relevant changed records were reviewed; no full atlas-delta stamp was made. Sparse checkout's earlier 0/0 ledger coverage is not a coverage finding. Atlas matching was run before proposing mechanisms; new questions below advance existing leads rather than claim novelty.

What is established, and what is missing

Track	Current evidence	Gap that matters next
Conditional hashed-ECDL	Compact encoder, exact counter, forced recursive controls, reduction/replay and strict inputs exist	Independent general proof-to-code and complexity audit; practical source guards select finite fallback
Finite hashed-ECDL	Selectors, last-slot derivation, exact width/ cutset plans, support propagation, partial relations and binary components implemented and development-tested	No admitted matched full-cost panel or evidence of an exponent improvement
Prime algebraic IC	PKM/tower machinery and preregistered fall-degree program exist	Naive prime-field reference arm is still a plan; solving degree and complete relation cost unresolved
Generic IC engineering	Table/pair/root-finding and setup improvements are implementation work	Compare against strongest matched rho; no algebraic exponent gain follows
Binary and transfer work	Frobenius-aware bases and special GHS-weak-family results exist	Preserve exact field/family, executable map, subgroup transport, useful relation rank and total cost
Execution/evidence	Historical checkpoint upload/readback recorded	That does not grant this session a native execution route or admit a new run

The canonical leaderboard says no row of tables A-C is below its reference. It also reports special GHS-weak-family wins outside those tables; those must not be generalized to arbitrary curves. Table C is 32-target batch work, not primary one-target evidence. Table B's prime IC/rho ratios 5.47-13.5 carry validated_with_caveats, including pricing_ok=False

and partly frozen protocol status. These are historical board entries, not new fully admitted measurements or a universal security conclusion. See the pinned board (https://github.com/isogeny-labs/crypto/blob/fc9af16fa1ce0654ee9df3ce63101c487ed215fb/docs/ic/LEADERBOARD.md).

The finite-size review's first five proposals have since received an implementation follow-up (../hashed_ecdl_finite_size_20261030/implementation/README.md). Do not propose their implementation again. Checked local plans certify finite conditioning/elimination, not a smaller-parameter Appendix-D packet transform. Partial-relation provenance exists; useful rank, yield, storage and total cost remain to be evaluated. Main's independent additions remain intact.

What changed in the external literature

Bochum v0.2 remains at 1ee44e56 (https://github.com/bochum-crypto/subexponential-ecdl/tree/1ee44e562b1b9a50ea339f3b8f81112b5a24b96f). It reports a conditional asymptotic ECDL reduction using general Boolean tensor contraction. Fixed cofactor and growing decomposition arity accompany its use. This is not a locally independently verified theorem or a concrete deployed- parameter attack.

Two primary formalization releases merit separate inspection: OpenAI's scope and linked development (https://github.com/openai/math/blob/4eaee87548250b2682affda7d05d947c05554bae/lean/docs/378.md) and Anthropic's ETH project (https://github.com/anthropics/formal-math/tree/16a5ce41304fcdaea4e0be7c132e1c361dc26da5/eth). Both publish counting/SAT statements; they are not interchangeable certificates for arbitrary succinct arithmetic predicates. This round read statement and entry files, not full proof dependency graphs, and did not run Lean.

The Hexagon weighted-counting extension (https://hexagonmath.org/2610.00594v1) reports local weights over commutative rational algebras. Our existing integration (../hexagon_cscc_20261030/README.md) distinguishes its exhaustive weighted reference from a port of that asymptotic engine. Ring, representation and coefficient-cost bridges remain research obligations.

Searches also returned quantum resource estimates and guess-and-determine claims. They were not advanced as classical hashed-ECDL leads: abstract/search listings do not discharge a classical complete-cost bound. This search is not evidence that no other method exists. No claim of novelty, consensus acceptance, impossibility or deployed-curve break follows.

Lead pursued now: theorem to code

Completed source finding: lean/formalization.yaml names OAI.Algorithms.SubexponentialSAT.SqrtLogSpace.exact_count_and_sat in SqrtLogSpace/Final.lean. Comparator JSON loads OAI.Algorithms.SubexponentialSAT.Main, which imports the actual solution. The comparator's sorry is an intentional statement placeholder, not a proof defect. The solution entry has proof terms; reading them is not kernel checking or an axiom audit.

The advertised endpoint counts fixed-width CNFs, with its exponent measured in CNF variables. Bochum needs a general implicit tensor theorem parameterized by network size, with specific entry-description and evaluation hypotheses. Arithmetic-circuit auxiliary variables must be charged. Polynomial-time predicate evaluation alone does not retain the quantitative saving; the source paper warns that naive low-arity expansion can erase it.

Completed native finding: Setup::new computes logarithmic size in usize; parameters divides it by 131072 before admitting recursion. On at-most 64-bit targets that scale is zero for every public representable graph. Counter::count selects Functional/Direct fallback or returns a resource error. Forced controls bypass admission and remain bounded correctness evidence. This known source boundary was checked again on current main; it does not refute an unbounded asymptotic theorem. No guard was changed.

BRIDGE.md (BRIDGE.md) separates source statements, formal endpoints, native invariants, entry work and acceptance criteria. It advances the audit from “read the proof” to named interfaces while leaving general validation open. The next source trace also found that newly created packet descriptions have no nested parent callback: they retain coefficient data and their own ring. The open cost target narrows to coefficient-extraction traversal, preparation, mode enumeration, retained data and aggregate bit work, rather than assuming callback composition is a present defect.

Next steps, ordered by the decision they resolve

Priority	Lead and next artifact	Discriminating criterion
1	General tensor bridge: pinned hypothesis/endpoint map, then declaration/dependency and bit-cost audit	Faithful general endpoint or quantitative reduction preserving the size saving; otherwise name the missing lemma
2	Finite solver discrimination: prepare paired complete-cost contract for delivered selectors, propagation and width plans	Charge decodes, supports, failures, slices, memory, useful rank and replay; continue on complete-cost improvement over frozen native baseline
3	Local packet admission: separate transform identities from sufficient asymptotic guards	Packing, every mode, creation precision, reconstruction, termination and total child/evaluator costs together; changing epsilon alone rejected
4	Prime algebraic reference: implement/price already specified Semaev-naive arm	Same instances, full solver cost, strongest matched rho and full pricing band; low first-fall degree alone insufficient

Priority	Lead and next artifact	Discriminating criterion
5	Shared membership coefficients: advance existing norm/quotient/Graeffe lead	Chart correspondence, full root/point recovery and preserved yield before comparing solve cost; excluded charts or hidden degree fail the shortcut
6	Binary/geometry: advance Frobenius and transfer obligations on exact fields	Replayable useful relation or executable subgroup-preserving transfer, then rank/cost; map or rank alone not a faster solver

Items 1, 3, 5 have source/proof work before paid computation. Item 2 prepares the existing measurement requirement; it does not turn test durations into speed claims. Existing approved jobs are not expanded by this assessment. No paid or scientific job was launched.

Fresh questions, without claiming new mechanisms

Typed predicate export. Native FnPredicate closures hide structure from solvers; width propagation enumerates their remaining support. Investigate a typed arithmetic/XOR representation with replayable propagation certificates. Preserve exceptional point cases and count multiplicities, and price building the representation. A cheaper answer from a different counted language is not a solver improvement. This advances existing SAT/encoding leads.

Complete plan cost before execution. A maximum-rank certificate does not price slice count, repeated tables, bit lengths and coefficient arithmetic. Can a source-bound per-instance cost certificate reject bad plans before solving? Compare to existing greedy plans only under an admitted protocol. This is a planning hypothesis, not an optimality certificate or measured gain.

Joint membership before independent root tests. Atlas lead LEAD-ICP-shared-coefficient-membership remains unexamined. Its norm/chart correspondence is a concrete proof task. Keep its empty squarefree-chart negative control, repeated-root arm and point recovery. It may expose a useful representation or only move degree cost elsewhere; neither outcome is a universal obstruction.

Weighted contraction as a separate bridge. Map rational coefficients and costs to the exact local ring/count interface first. Do not substitute min-plus or approximate truncation for exact counting. Existing exhaustive profiles are a reference, not the fast engine.

Immediate continuation and boundaries

First lead pursued: BRIDGE.md (BRIDGE.md). Next inspect general implicit-entry recursive description/evaluation work against immutable creation-precision callbacks. Preserve a first counterexample if found; do not lower guards to make an example run. The general-audit problem (../../problems/hashed-ecdl-tensor-backend.md) stays open.

Verified this round: current pins; delivered/planned distinction; relevant atlas collisions and current metadata counts; formal endpoint tracing; native admission source review; scoped report and visual validation. Partial: general proof-to-code audit and broad program coverage. Not done: kernel verification, new experiments, timing, challenge solves, novelty certification, new transfers or scientific state promotions.

No new measured cost, curve or admitted bound: canonical leaderboard, progress-timeline, index-calculus scoreboard and curve/performance graphs stay unchanged. The diagram is explanatory. Exact curve/subgroup identities for a later run must come from its frozen protocol; this assessment targets no key or live instance. Old root/setup receipts remain source-specific historical validation, not tests of current main.

Conductor is absent (127); documented warn-and-allow fallback, no reservation. Sparse-file reads were repaired by materializing tracked paths. A guessed Lean URL returned 404; published scope/catalog links located the actual solution. An initial patch rejected duplicate delete/add operations before applying; rewrote only the unpublished draft after saving a scratch copy. Earlier draft wording prematurely claimed a PDF/diagram; those artifacts are checked before publication here. No frozen receipt, manifest, dependency or gate was edited.

First pursued lead: general contraction to ECDLP

Source inspection, 2026-10-30. No independent kernel check, proof of the general theorem, new mathematical identity or correctness theorem is claimed. Read sources.json (sources.json) with this map.

Interface	Source and observation	Acceptance still needed
ECDL reduction	Bochum v0.2 Theorem 2 imports OpenAI D.1; Theorem 4 uses growing arity and fixed cofactor	Exact general tensor theorem, not merely a SAT consequence
OpenAI routing	formalization.yaml lists Final.lean:exact_count_and_sat; comparator loads SubexponentialSAT.Main, importing Final	Independently build pinned dependencies and audit declarations/axioms
OpenAI endpoint	Fixed-width CNF counting, primitive-machine time and all-prefix visited space	General implicit-entry endpoint or quantitative reduction preserving the ECDL size saving
Comparator semantics	Challenge statement has a placeholder; solution has proof terms; config lists standard permitted axioms and enable_nanoda=false	Placeholder is not a missing solution; no local kernel result obtained
Alternate source	Anthropic MainResults defines ETHIsFalse/ MainTheorem; Solution provides terms via CountedRoute, NamesInOrder and Wrap	Independent source/cost checking and an actual engine port; not already the Rust backend
Entry work	D.4.6 retains immutable creation descriptions/precision and shares references	General recursive entry-work and retained-space bound against native callbacks
Native admission	Setup::new/parameters and Counter::count use finite fallback at public machine sizes	Distinguish forced controls from an admitted scalable execution; preserve guards
Finite plans	width.rs:Plan::validate rechecks pins, partition and intermediate ranks	Rank cap is not a smaller-packet proof or complete runtime estimate
Encoding	encoding.rs uses implicit closures and arithmetic metadata	Uniform degree/wire/description/ construction bit-work bounds, including pins and exceptional cases
Sampling	Counter::sample documents completion weighting; selector counts have their own contract	Track projected words, cofactor preimages, auxiliary witnesses and seed distribution
Whole reduction	Remark 2 separately prices oracle calls, elimination and memory	Retain all terms, failed attempts and rank/ invertibility probability

Avoid false shortcuts

Fixed-k formal counting is not the same endpoint as arbitrary implicit tensor contraction. Its exponent uses encoded variables. A circuit simulation may introduce enough auxiliary variables to lose the intended saving; the missing item is the appropriate quantitative bound, not mere computability.

Native fallback is a representation boundary, not a counterexample to an asymptotic existence theorem. Passing fallback tests does not verify recursion. Generic selector enumeration and coordinate-aware tensor predicates also have different computational models; coordinate access alone proves no advantage.

Readable proof terms, listed axioms and source hashes do not replace checking the pinned dependencies. No Lean executable was present in this environment.

Next packet: implicit-entry cost

Additional source trace completed in this round

Tensor::entry charges an entry, checks creation precision and, for a packet, evaluates packet.entry in its stored ring before lifting. PacketTensor contains its ring, local coefficient table and incoming/outgoing mode data; it has no parent tensor callback. child retains old vertices by Arc and constructs new packet data from the Cauchy kernel and selected modes. Thus the suspected *nested parent callback inside each newly created packet* is not present in this representation. This is a source finding, not a general complexity proof or an assertion that all descriptions are cheap.

The concrete remaining cost interface is coefficient extraction: PacketTensor::entry performs a depth-first choice traversal over rows and incoming exponents. evaluate_at enumerates weighted mode tuples and group children; strict mode checks edge/degree shrinkage. Audit the total traversal, coefficient/table preparation, retained data and bit work together. Meter charges bound an admitted execution but do not independently prove the source's uniform asymptotic work bound. Original Boolean predicates also need the source's evaluation-cost hypothesis; an opaque user closure does not supply it by its type alone.

Follow one retained tensor and one new packet tensor through creation, lift, child evaluation and reconstruction. Record parent-description references, creation precision, port count, coefficient bit size, temporary storage and predecessor evaluations. Compare the source entry-work induction with native callbacks; include weighted-mode enumeration and depth-first accumulation.

Success requires a complete invariant with checked dependencies, or an exact missing assertion/counterexample. A bounded control is regression evidence, not the general induction. New algebraic results require the repository's identity/proof rules; this map records existing obligations only.

Cost record before a later run

Bind revision, encoding/backend, curve/subgroup, seeds and protocol. Price setup, decoding, support checks, predicates, unsuccessful queries, useful rank, elimination, memory and verification. Keep Limit/timeout unknown. Compare unchanged workloads and independently replayed answers against the frozen native baseline and strongest matched rho. Do not combine historical units, compare reused setup with a cold reference, or claim fastest without external comparisons.